

[Soal] Project Trinity

AZZAM LABIB HAKIM | INSTAGRAM: @HAXUV_WORLD | THREADS: @AZZAM_29_12

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§1 Soal

§1.1 KTOM Juli 2023 Simulasi Pra OSN

Tentukan semua tripel bilangan asli (a, b, c) sehingga ketiga bilangan berikut:

$$2^a + 2^b + 3(c + 1), \quad 2^b + 2^c + 3(a + 1), \quad 2^c + 2^a + 3(b + 1)$$

semuanya adalah perpangkatan dua (dengan kata lain, dalam bentuk 2^k untuk suatu bilangan asli k).

(KTOM Juli 2023 Simulasi Pra OSN)

§1.2 Semifinal PSN IPB 2023

Diberikan $a, b, c \in \mathbb{R}^+$. Jika $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 3$, buktikan bahwa

$$\frac{1}{\sqrt{a^3 + b}} + \frac{1}{\sqrt{b^3 + c}} + \frac{1}{\sqrt{c^3 + a}} \leq \frac{3}{\sqrt{2}}.$$

(Semifinal PSN IPB 2023)

§1.3 Random 2 AMO Exercise

Sebuah bilangan palindrom 6 digit dengan digit terakhir 4 merupakan hasil perkalian antara dua atau lebih bilangan asli berurutan. Hitunglah hasil penjumlahan digit-digit palindrom tersebut.

§1.4 All Russian Olympiad 1998 Problem 6 Grade 10

A binary operation $*$ on real numbers has the property that $(a * b) * c = a + b + c$ for all a, b, c . Prove that $a * b = a + b$. (All Russian Olympiad 1998 Problem 6 Grade 10)

§1.5 USAJMO 2024 Problem 5

Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ that satisfy

$$f(x^2 - y) + 2yf(x) = f(f(x)) + f(y)$$

for all $x, y \in \mathbb{R}$.

(USAJMO 2024 Problem 5, Proposed by Carl Schildkraut)

§1.6 Random 3

Terdapat tepat satu bilangan real a sehingga persamaan

$$4 \lfloor ax \rfloor = x + \lfloor a \lfloor ax \rfloor \rfloor$$

berlaku untuk sembarang bilangan asli x . Carilah nilai $\lfloor a \rfloor$.

§1.7 Random 4

Fungsi $f : \mathbb{R} \rightarrow \mathbb{R}$ memenuhi

$$f(f(x)) = x^2 - x + 1$$

untuk semua bilangan real x . Tentukan nilai $f(0)$.

<https://x.com/mathssolutionz/status/1795132058475397555>

§1.8 Random 5

Bilangan bulat positif x dan y memenuhi persamaan

$$x - y - \frac{x}{y} + \frac{x^5}{y^5} = 2024.$$

Tentukan nilai x terkecil yang memenuhi.

§1.9 Random 6 Quick Inequality

Misalkan $a, b, c > 0$ dan memenuhi $a + b + c = 1$. Tunjukkan bahwa

$$\frac{ab}{1+c} + \frac{bc}{1+a} + \frac{ca}{1+b} \leq \frac{1}{4}.$$

§1.10 Fungsi Bulat 0

Jika $f : \mathbb{Z} \rightarrow \mathbb{Z}$ merupakan fungsi tak terbatas, dan $f(a) \mid c$ untuk konstanta c dan $a \in T \subseteq \mathbb{Z}$.

Buktikan jika $|T| = \infty$ maka $c = 0$.

§1.11 Fungsi Bulat 1

Tentukan semua fungsi $f : \mathbb{N} \rightarrow \mathbb{N}$ sehingga untuk setiap bilangan bilangan asli a, b berlaku:

$$f(a) + b \mid (f(b) + a)^2$$

§1.12 Fungsi Bulat 2

Tentukan semua $f : \mathbb{N} \rightarrow \mathbb{N}$ yang memenuhi

$$n + f(m) \mid f(n) + nf(m)$$

untuk semua $m, n \in \mathbb{N}$

§1.13 Fungsi Bulat 3

Tentukan semua fungsi $f : \mathbb{N} \rightarrow \mathbb{N}$ yang memenuhi:

$$f(a) + f(b) \mid a + b, \forall a, b \in \mathbb{N}$$

§1.14 Baltic Way 2012 Problem 3

(a) Show that the equation

$$\lfloor x \rfloor (x^2 + 1) = x^3,$$

where $\lfloor x \rfloor$ denotes the largest integer not larger than x , has exactly one real solution in each interval between consecutive positive integers.

(b) Show that none of the positive real solutions of this equation is rational.

(Baltic Way 2012 Problem 3)

§1.15 OSK 2019 Problem 4

Misalkan $f(x) = 1 + \frac{90}{x}$. Nilai terbesar x yang memenuhi

$$f(f(\cdots f(x) \cdots)) = x,$$

(2019 kali aplikasi fungsi f) adalah ...

(OSK 2019 Nomor 4)

§1.16 Random 7 Functional Equation

Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ which satisfy

$$f(x+y) + f(x)f(y) = f(x) + f(y) + f(xy)$$

for all $x, y \in \mathbb{R}$.

§1.17 SMO 2024 Junior Round 1

English Let a , b and c be real numbers such that $a + b + c = 8$ and $ab + bc + ca = 0$. Find the maximum value of $3(a + b)$.

Bahasa Indonesia Misalkan a , b dan c adalah bilangan real yang memenuhi $a + b + c = 8$ dan $ab + bc + ca = 0$. Tentukan nilai maksimum dari $3(a + b)$.

§1.18 Random 8 Inequality

Prove that for any positive real numbers a, b, c it applies

$$\frac{(a+b)^2}{c^2+ab} + \frac{(b+c)^2}{a^2+bc} + \frac{(c+a)^2}{b^2+ac} \geq 6$$

§1.19 Random 9 Inequality

a, b, c are non-negative real numbers for which holds that $a + b + c = 3$. Prove the following inequality:

$$4 \geq a^2b + b^2c + c^2a + abc$$

§1.20 OSP 2019 Esai Problem 2 - System of Equalities

Cari semua bilangan real k sehingga sistem persamaan

$$\begin{cases} a^2 + ab = kb^2 \\ b^2 + bc = kc^2 \\ c^2 + ca = ka^2 \end{cases}$$

memiliki solusi bilangan real positif a, b, c .

§1.21 Random 10 Functional Equation

Find all functions f where

$$(x+y)(f(x)-f(y)) = f(x^2) - f(y^2)$$

for all real numbers x, y .

§1.22 Random 11 Combinatorics Set Counting

Diketahui A, B, C adalah himpunan bagian dari $\{1, 2, 3, 4\}$. Tentukan banyaknya tripel himpunan (A, B, C) yang memenuhi

$$A \subseteq B \quad \text{dan} \quad A \cap C \neq \emptyset$$

§1.23 Canadian MO 2013 Polynomial

Tentukan semua polinomial $P(x)$ dengan koefisien real sedemikian sehingga

$$(x+1)P(x-1) - (x-1)P(x)$$

adalah polinomial konstan.

§1.24 HMMT 2017 Polynomial

Misalkan $P(x)$ dan $Q(x)$ adalah polinomial tak konstan dengan koefisien-koefisien real. Buktikan bahwa jika

$$\lfloor P(y) \rfloor = \lfloor Q(y) \rfloor$$

untuk semua bilangan real y , maka $P(x) = Q(x)$ untuk semua bilangan real x .

§1.25 SMO 2025 Open Problem 3

For any 4-digit positive integer n , define $f(n) = (a + b)^2$, where a, b are the numbers formed by the first two and last two digits of n , respectively (leading zeros are allowed). For instance,

$$f(2025) = (20 + 25)^2 = 45^2 = 2025.$$

Find all 4-digit positive integers n such that $f(n) = n$.

§1.26 ARMO 2013 Grade 11 Problem 6

Let a, b, c, d be positive real numbers such that $2(a + b + c + d) \geq abcd$. Prove that

$$a^2 + b^2 + c^2 + d^2 \geq abcd$$

§1.27 OSP 2025 Isian Singkat Problem 4

Misalkan bilangan asli a, b, c, d memenuhi persamaan

$$2^a + 2^b + 2^c = 4^d.$$

Jika $a + b + c + d \leq 500$, maka nilai terbesar yang mungkin dari d adalah...

§1.28 EMC Junior 2024 Problem 1

Wiske wrote a 2024-digit positive integer on the blackboard. In each round of the game she erases the last digit of the integer, let this digit be d , and writes down the sum of the remaining number and $2d$ in place of the old number. She repeats the same steps with the newly obtained number. After a certain number of rounds, Wiske found that the new number obtained was the same as the number in the last round and she stopped the game. What is the smallest possible 2024-digit integer that Wiske started with in this game?

(Proposed by Kai Chen)

§1.29 Random 12 Three Variables Inequality

Let a, b, c be positive real numbers. Prove the inequality

$$\frac{a^2}{b} + \frac{b^2}{c} + \frac{c^2}{a} \geq a + b + c + \frac{4(a-b)^2}{a+b+c}.$$

§1.30 Classical Probability of n Points in a Semicircle

Choose n points randomly on a circle. What is the probability that all points lie in a semicircle?

Terjemahan: Pilih n titik secara acak pada sebuah lingkaran. Berapakah probabilitas bahwa semua titik tersebut terletak dalam sebuah setengah lingkaran?

§1.31 Random 13 Four Variables Inequality

Let a, b, c, d be positive real numbers such that $abcd = 1$. Prove the inequality

$$\frac{ab+1}{a+1} + \frac{bc+1}{b+1} + \frac{cd+1}{c+1} + \frac{da+1}{d+1} \geq 4.$$